



Machine Learning Basics

CS-4025: Deep Learning for Perception

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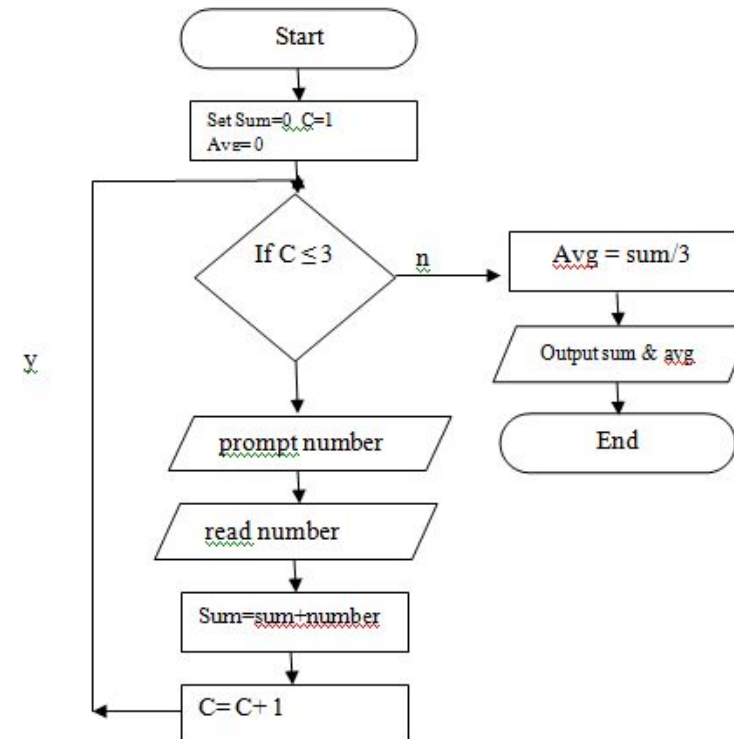
Department of Computer Science

Goals

- What is learning?
- What is machine learning?
- Types of machine learning
 - Classification vs Regression

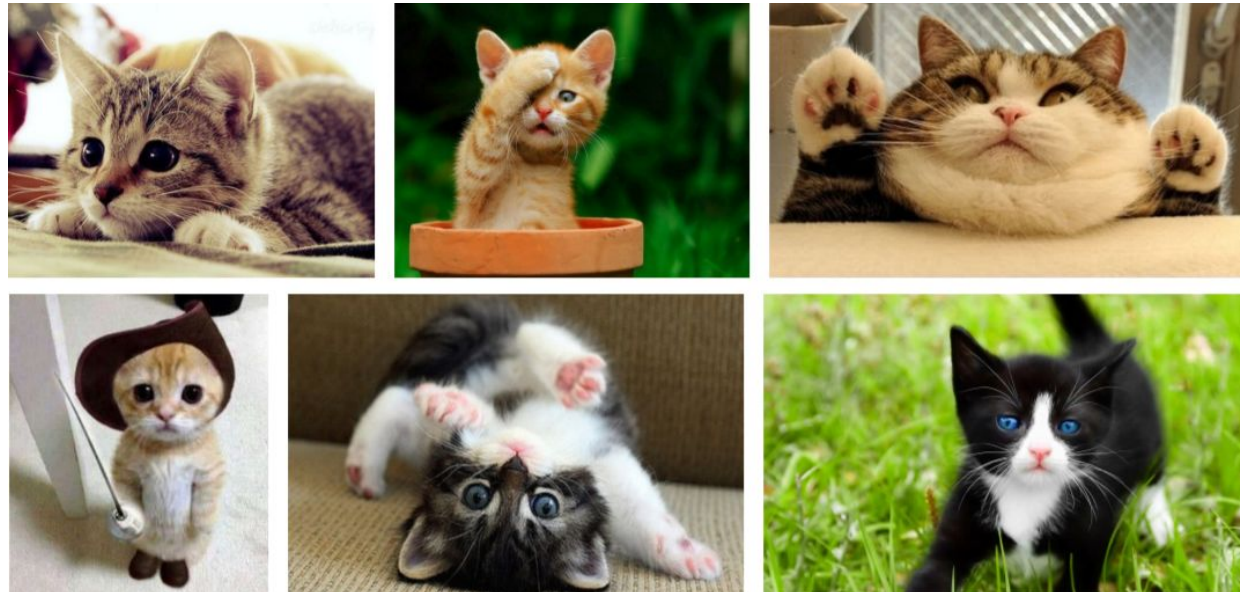
What is Learning?

- How can we solve a specific problem?
 - We write a program with a **set of rules** that are useful to solve the problem.
 - **Example**: Find average of three numbers



What is Learning?

- In many situations it is very difficult to **specify those rules** to solve a problem.
- For example, given a picture determine whether there is **a cat in the image**



What is Learning?

- Find face of a specific person?



What is Learning?

Benign vs Malignant tumor

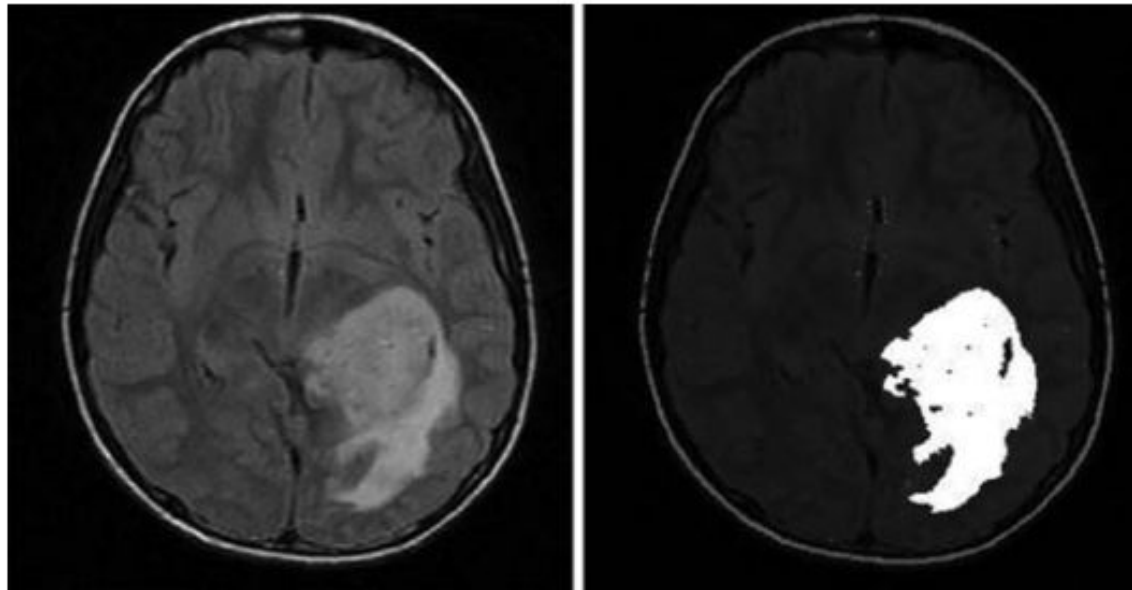
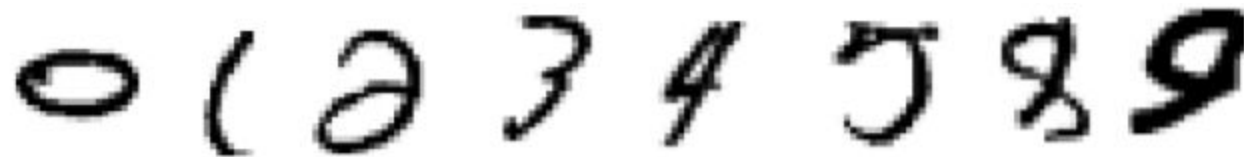


Figure 2. Gradient based genetic algorithm: (i) Original MRI image(ii) Brain tumor segmentation (KumarKole et al., 2012).

What is Learning?

- Any learning systems are not directly programmed using conditions to solve a problem
- Instead it should learn from examples (data)
- From trial-and-error experience trying to solve the problem



What is Machine Learning?

- Machine Learning is the **science (and art)** of programming computers so they can *learn from data*.
- *[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.*
 - Arthur Samuel, 1959

What is Machine Learning?

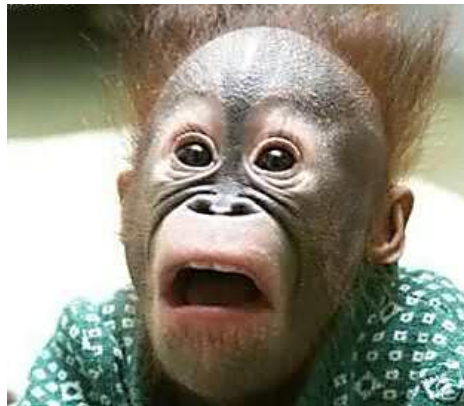
- Machine learning can be defined as computational methods using **experience** to **improve performance** or to **make accurate predictions**.
- *Experience* refers to the **past information**.

Mohri et al

What is Machine Learning?

- Definition: “A computer program is said to learn from **experience E** with respect to some class of **tasks T** and **performance measure P**, if its performance at tasks in T, as measured by P, improves with experience E”

Tom M. Mitchel



A checkers learning problem

- **Task T:** playing checkers
- **Performance measure P:** percent of games won against opponents
- **Training experience E:** playing practice games



Spam Tagging Problem

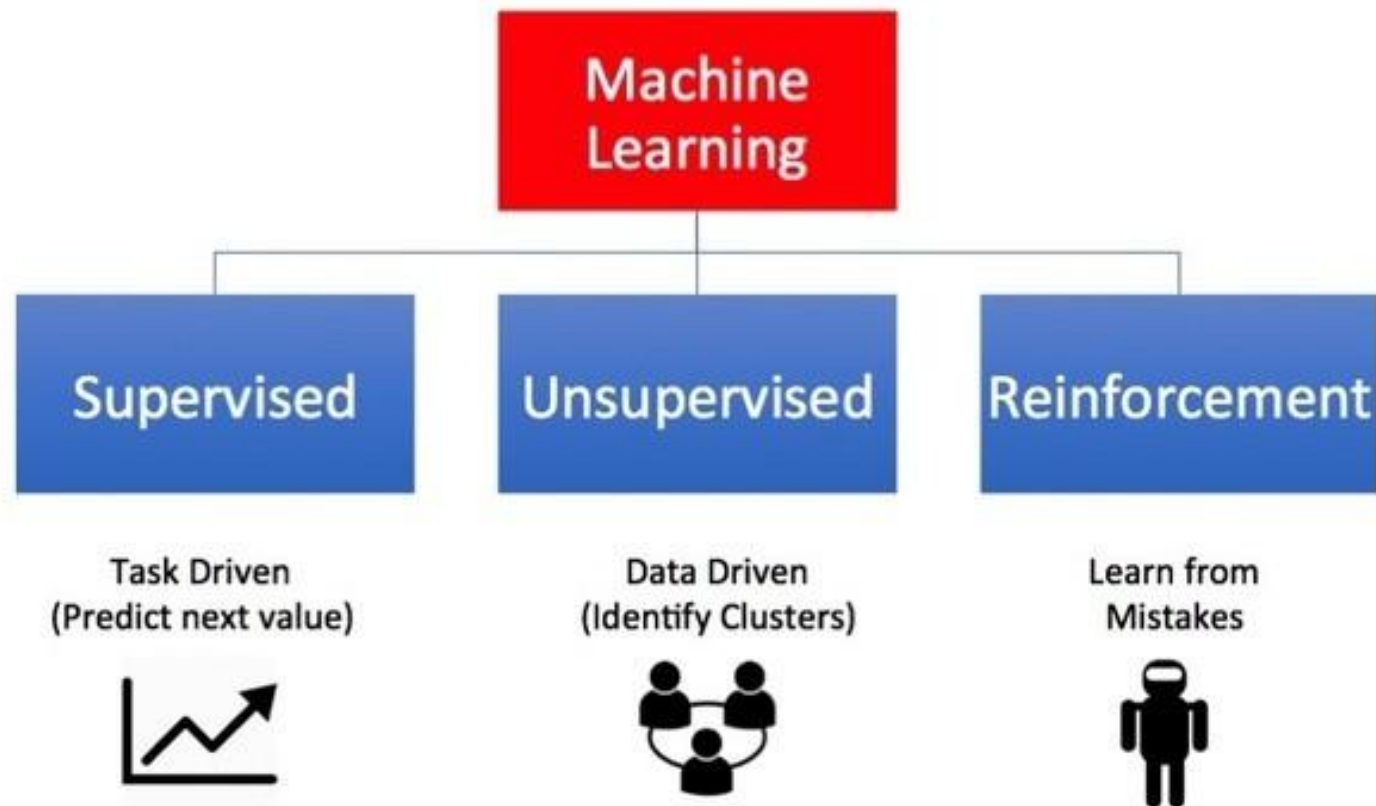
- Your **spam filter** is a Machine Learning program
 - **Binary Classification Problem**: spam emails or nonspam
- To train a machine learning model, examples of emails that are **spam and nonspam** should be presented to the model
 - Usually flagged by users
- The examples that the model uses to learn are called the **training set**.
 - Training instance (or sample).

Spam Tagging Problem

- For Spam classification:
 - The **task T** is to flag spam for new emails
 - The **experience E** is the *training data*,
 - The **performance measure P** needs to be defined;
 - Percentage of correctly classified emails (*accuracy*)

Types of Machine Learning...

Types of Machine Learning



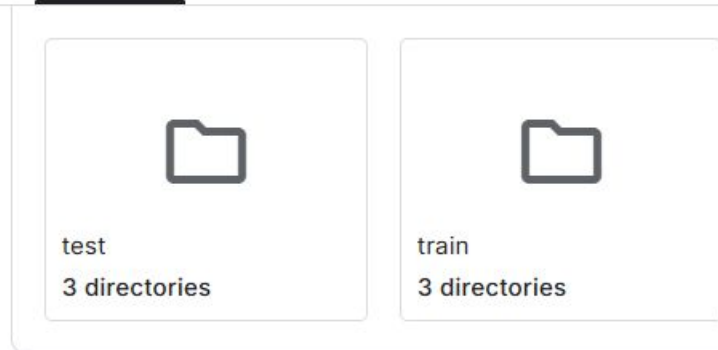
Supervised learning

- For supervised learning, we provide **both data and labels** for training the algorithm.
- The algorithms learns from the **data and labels**
- After training, we can pass **test samples** to check if the **algorithm learned the data or not**
- **Most popular** in ML community

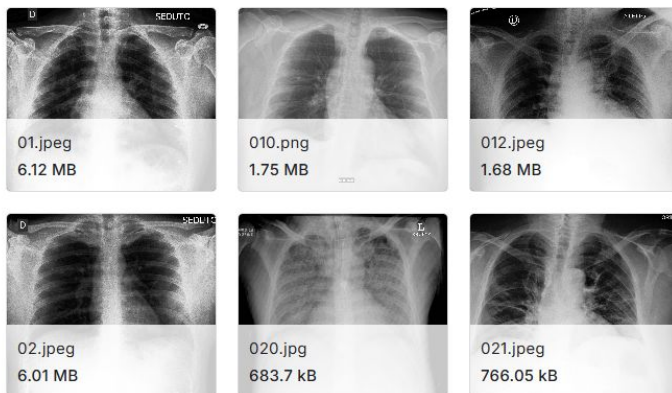
Supervised learning: Example

Covid-19 Image Dataset

Data Card Code (157) Discussion (3) Suggestions (0)

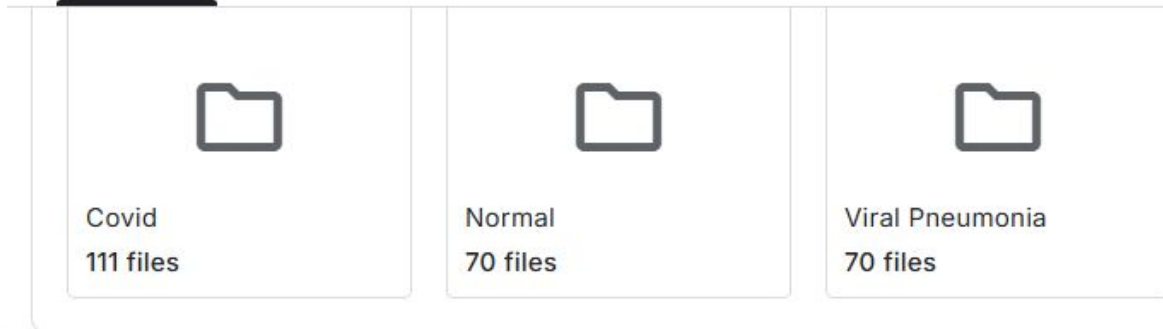


Covid (111 files)

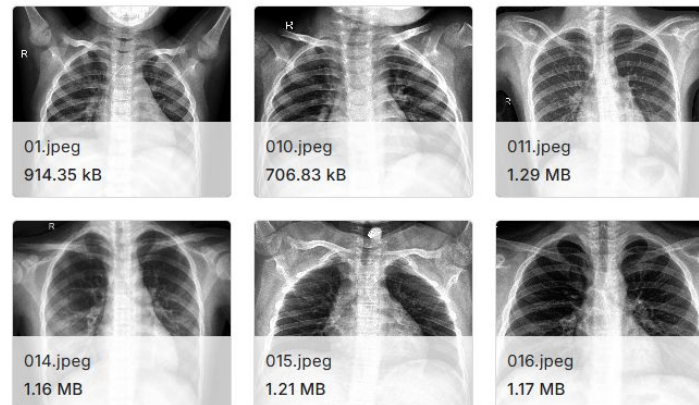


Covid-19 Image Dataset

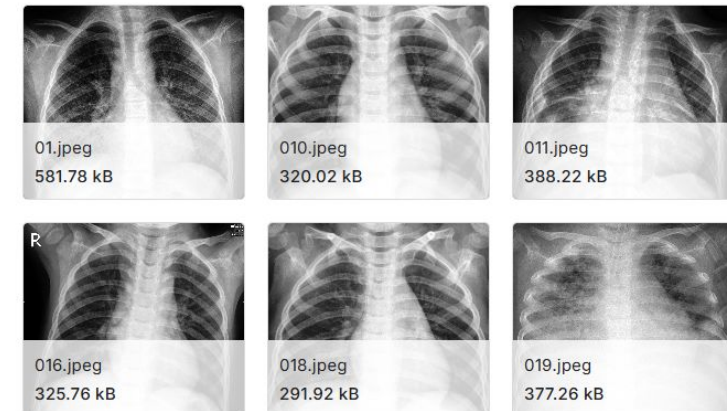
Data Card Code (157) Discussion (3) Suggestions (0)



Normal (70 files)



Viral Pneumonia (70 files)

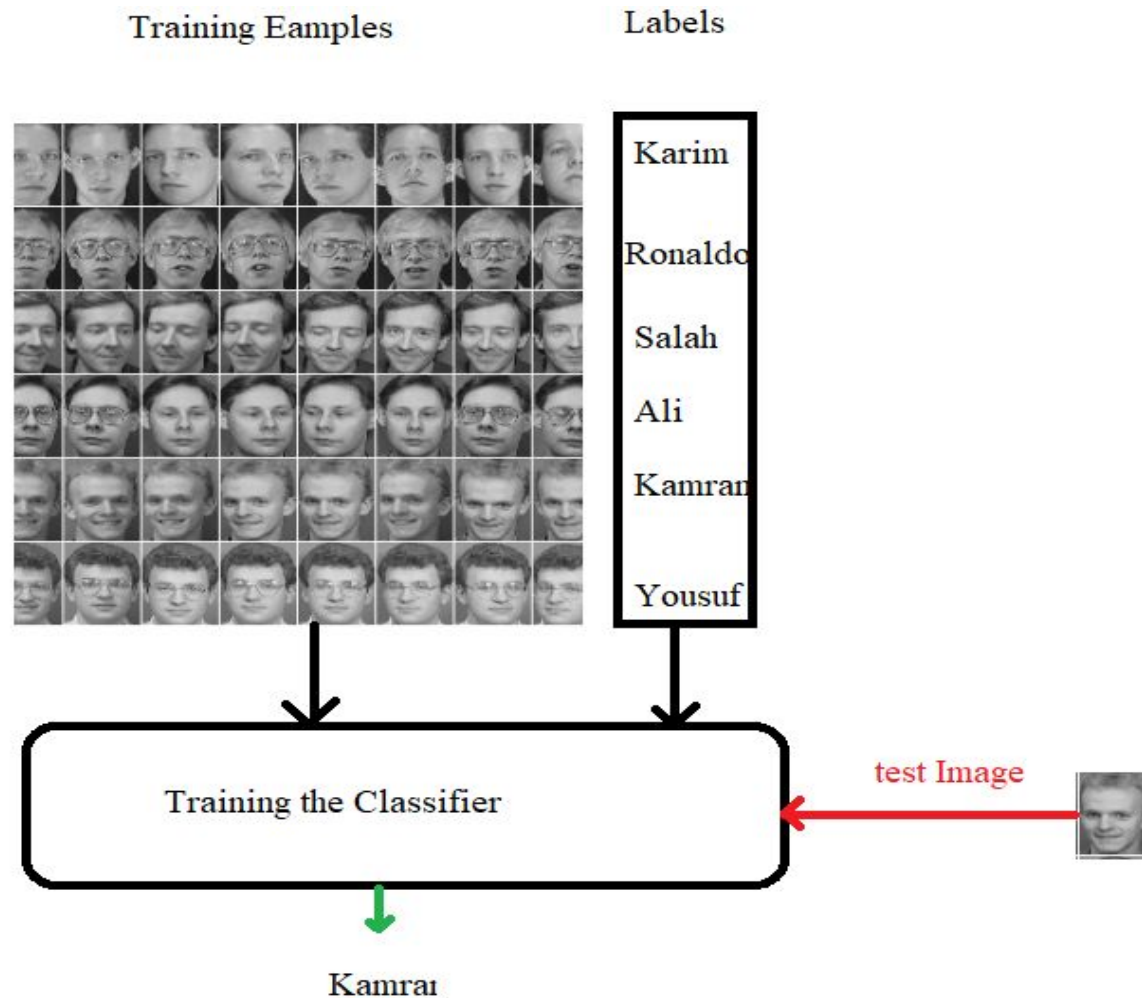


Data Explorer

Version 2 (169.64 MB)

- ▼ Covid19-dataset
 - ▼ test
 - ▶ Covid
 - ▶ Normal
 - ▶ Viral Pneumonia
 - ▼ train
 - ▶ Covid
 - ▶ Normal
 - ▶ Viral Pneumonia

Supervised learning: Example



Supervised learning: Example

- Supervised learning is widely used for tasks such as:
- **Classifying medical images.**
 - Given a large datasets of images of **malignant and benign tumors**.
- **Translating between pairs of languages.**
 - Pairs of sentences that have the same meaning in **two different languages**.
- **Detecting objects in a self-driving car.**
 - Provide the model with examples of **cars, pedestrians**, etc.

Supervised Learning

Feature Space $\mathcal{X}$

Words in a document

Label Space $\mathcal{Y}$

"Sports"
"News"
"Science"
...



Cell properties

"Anemic cell"
"Healthy cell"



Discrete Labels

Supervised learning

Data: $X = \{x_1, x_2, \dots, x_n\}$ **n examples**

$$d_i = \langle \mathbf{x}_i, y_i \rangle$$

$\mathbf{x}_i$ is input vector, and y is desired output (given by a teacher)

Objective: learn the mapping $f : X \rightarrow Y$

s.t. $y_i \approx f(x_i)$ for all $i = 1, \dots, n$

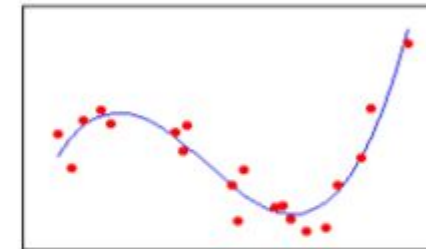
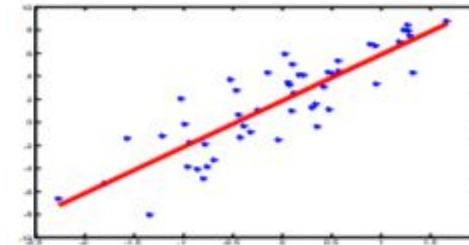
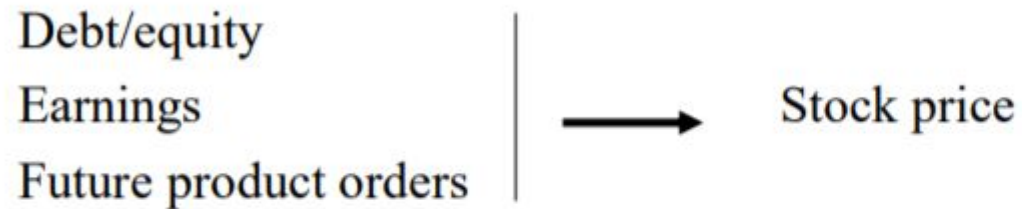
Supervised learning

Two types of problems:

- **Regression:** X discrete or continuous $\rightarrow$
Y is **continuous**
- **Classification:** X discrete or continuous $\rightarrow$
Y is **discrete**

Supervised learning

- **Regression:** Y is **continuous**



Data:

Debt/equity	Earnings	Future prod orders	Stock price
20	115	20	123.45
18	120	31	140.56
....			

Supervised learning

- **Classification:** Y is **discrete**



Handwritten digit (array of 0,1s)



Data:



image



digit

3

7

5

Supervised learning

- Can regression algorithms be used for classification and vice versa?
 - Yes, some algorithms can be used.
- Logistic Regression is commonly used for classification
 - Predicts probability belonging to a class

Supervised learning

- Some widely used supervised ML algorithms:
 - Linear Regression
 - Logistic Regression
 - Support Vector Machines (SVMs)
 - Decision Trees and Random Forests
 - Neural networks
 - k-Nearest Neighbors

Unsupervised learning

- For unsupervised learning, we provide data but NOT labels for training the algorithm
- The system tries to learn without a teacher.
- Learns relations among data by itself
- Then put the data into different groups/clusters

Unsupervised Learning

What is a natural grouping?



Clustering is subjective



Simpson's Family



School Employees

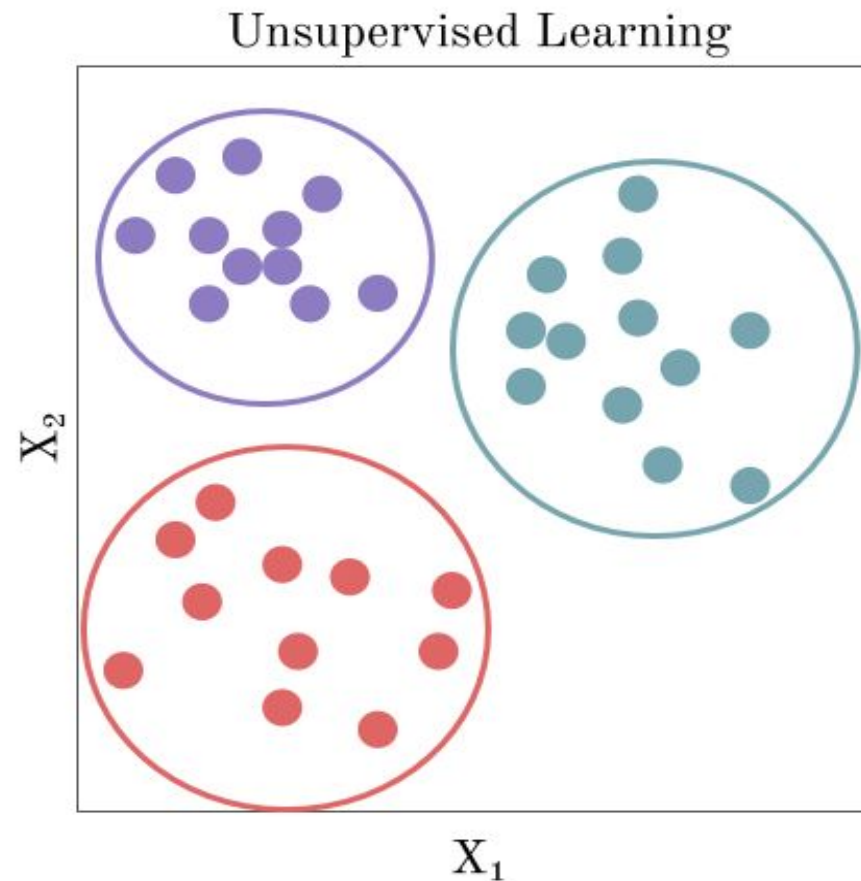


Females



Males

Unsupervised Learning



Unsupervised Learning

- Some widely used unsupervised learning algorithms:
 - K-Means
 - Principal Component Analysis (PCA)
 - Apriori
 - Hierarchical Cluster Analysis (HCA)
 - One-class SVM

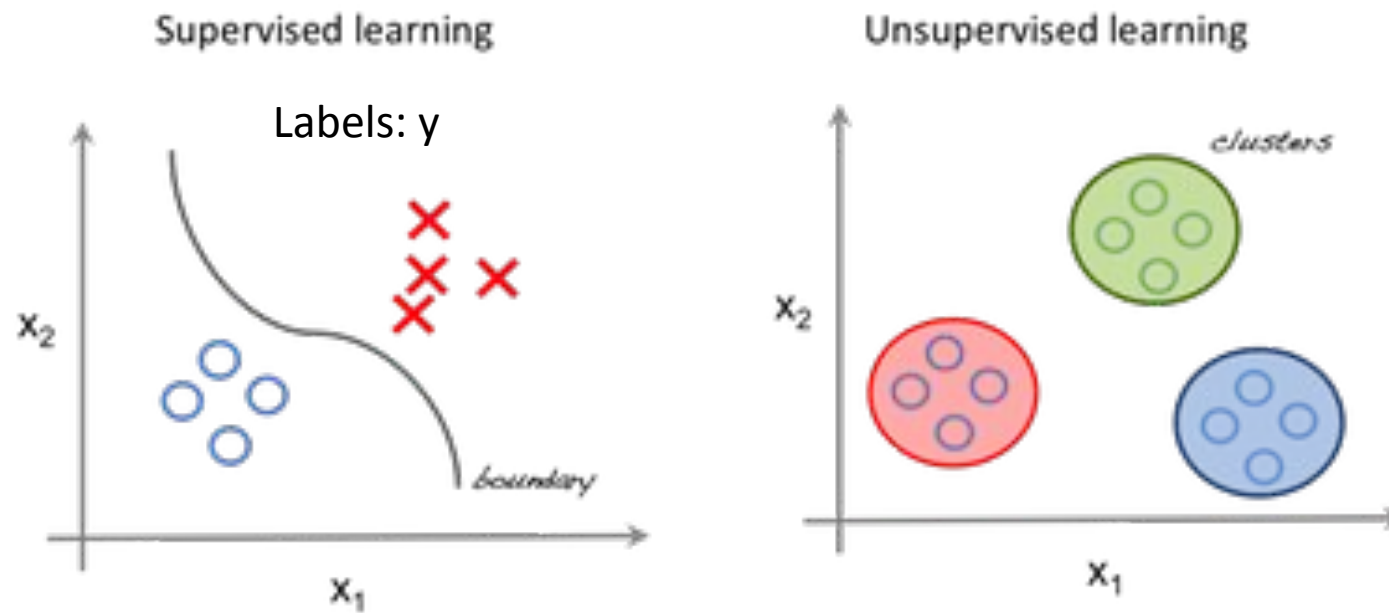
Usage of Unsupervised Learning

- Data visualization
- Dimensionality reduction
- Clustering
- Anomaly detection

Unsupervised Learning

- **Data:** $x = \{x_1, x_2, \dots, x_n\}$ vector of values
No target value (output) y
- **Objective:**
 - learn relations between samples, components of samples

Supervised vs Unsupervised Learning



Reinforcement learning

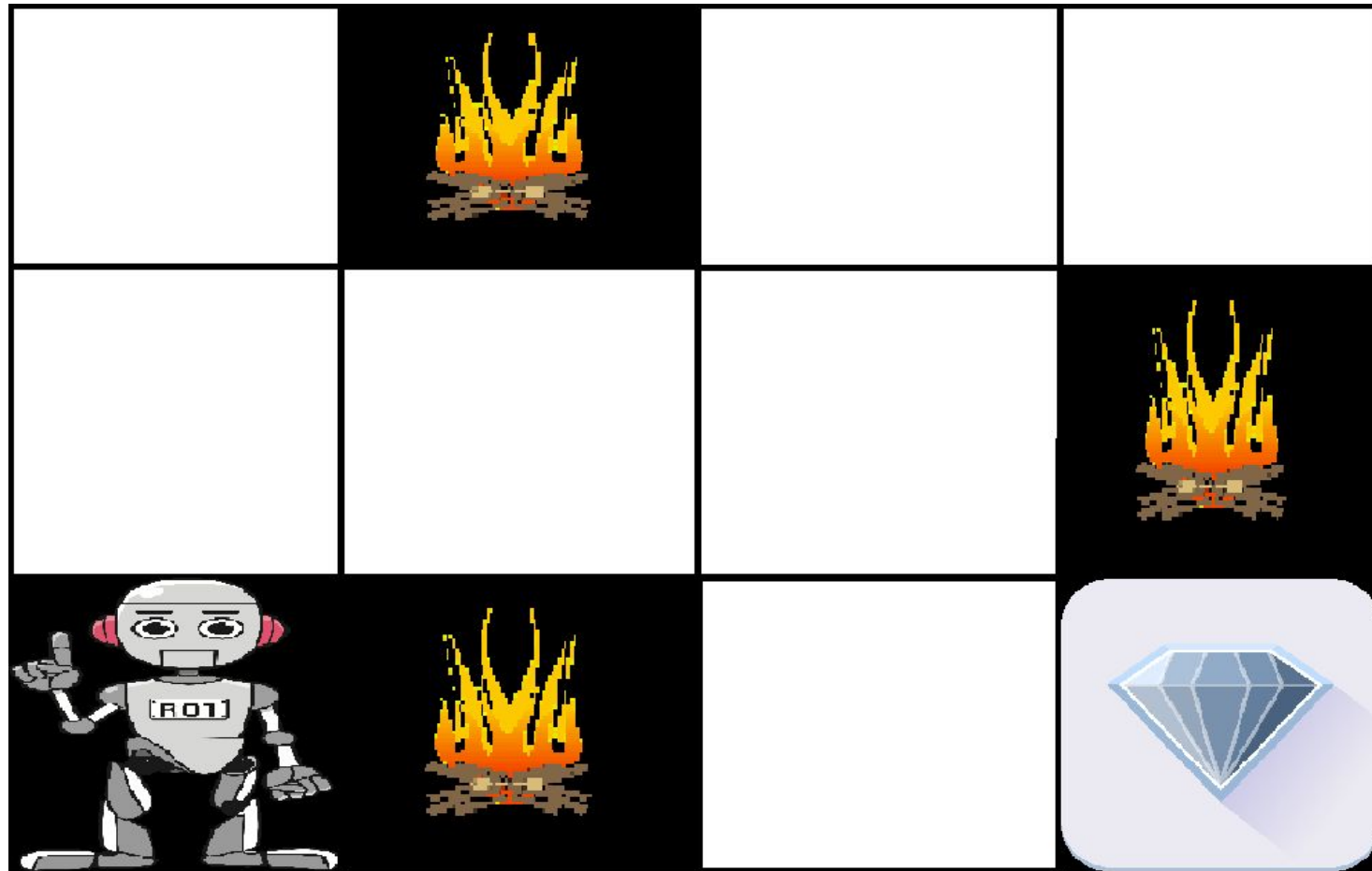
- The learning system, called an **agent**, can **observe the environment, select and perform actions**:
 - Get **positive rewards** for good actions
 - Get **negative rewards** for wrong action
- Reinforcement learning **refers to goal-oriented algorithms**, which learn how to attain a complex **objective (goal) or maximize**

Reinforcement learning

- It must then learn by itself what is the best strategy
 - **Policy**: best strategy
- A policy defines what **action the agent should choose** when it is in a given situation.
- Example:
 - Playing games, Robotics
 - Robots learn how to walk.
 - DeepMind's AlphaGo



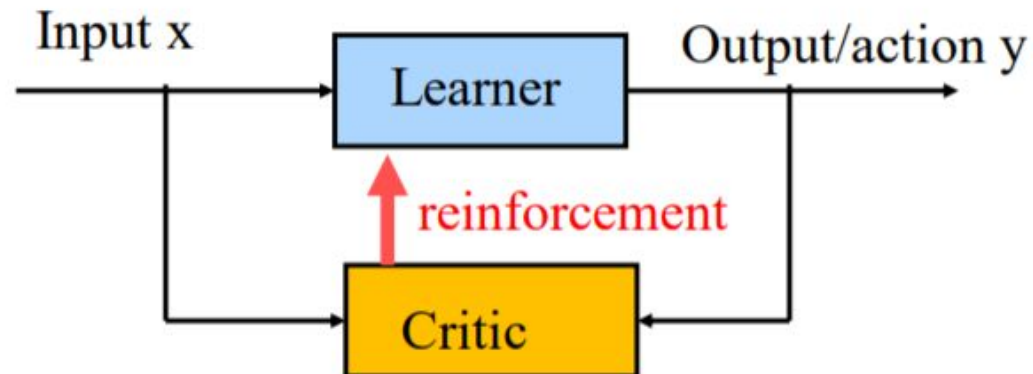
Reinforcement learning



Reinforcement learning

We want to learn: $f : X \rightarrow Y$

- We see examples of inputs x but not y
- We select y for observed x from available choices
- We get a feedback (reinforcement) from a **critic** about how good our choice of y was



- The goal is to select outputs that lead to the best reinforcement

Curiosity: Question of the Day

- What if the **data is changing**, should we **retrain** the model from scratch?
- Or anything else can be done?

Reference

- Read 1st Chapter of Hands-on Machine Learning with Scikit-Learn, Keras, and TensorFlow- (2019)
- <https://towardsdatascience.com/workflow-of-a-machine-learning-project-ec1dba419b94>
- <https://cloud.google.com/ai-platform/docs/ml-solutions-overview>

Thank You 😊